Closed Mapping

Def:

A map

$$f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$$

is called a **closed map** iff the image of each closed subset of X is closed in Y.

$$\forall\, U \underset{\text{closed}}{\subset} X \Rightarrow f(U) \underset{\text{closed}}{\subset} Y$$

Projection Mapping

Fact: mcQ

Projection map may or may not be closed

Justification:

Consider the first projection

$$\pi_1 : (\mathbb{R} \times \mathbb{R}, \mathcal{T}_{\mathbb{R}\times\mathbb{R}}) \to (\mathbb{R}, \mathcal{T}_u)$$

The set $A = \{(x, y) \mid y = \tan x, \frac{-\pi}{2} < x < \frac{\pi}{2}\}$ is closed in $\mathbb{R} \times \mathbb{R}$ but

$$\pi_1(A) = \boxed{\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)}$$

Not close

VU

Projection Mapping

Projection Map:
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces. The maps

$$\pi_1 : (X \times Y, \mathcal{T}_{X \times Y}) \to (X, \mathcal{T}_X)$$
$$(x, y) \mapsto x$$

$$\pi_2 : (X \times Y, \mathcal{T}_{X \times Y}) \to (Y, \mathcal{T}_Y)$$
$$(x, y) \mapsto y$$

are called projection maps, where $\mathcal{T}_{X \times Y}$ is product topology on $X \times Y$. π_1 is first projection and π_2 is second projection.

VU

Openness and Restriction...

Now f is an open map but f_A is not an open map. Because $\{b\}$ is an open subset of A but

$$f_A(\{b\}) = \{2\} \underset{Not\,Open}{\subset} Y$$

Remark:

MCQs

Restriction of an open map may or may not be an open map.

nd Open Mapping

MCQs

Answers:

1. Continuous map may or may not be open map.
2. Open map may or may not be continuous map.

Homeomorphism...

define

Homeomorphism:
A map $f : X \to Y$ is called a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. f is bijective.
2. f is continuous.
3. $f^{-1} : Y \to X$ is continuous.

Homeomorphism...

define **Homeomorphic Spaces:** Two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are called homeomorphic iff there exists a homeomorphism $f : X \to Y$. We also say such spaces topologically equivalent spaces.

opological Properties

Define **Topological Property**:

A property "P" of a topological space (X, T) is called "Topological Property" if every space homeomorphic to X has property P.

i.e.

A property that is unchanged under homeomorphism.

Topological property is also called topological invariant.

S. question

Proposition:

Show Homeomorphism of topological spaces is an equivalence relation $(\approx)$

✓ Proof: 1:. $X \approx X$ $\quad Id: X \to X$
$x \mapsto x$

2:. Let $X \approx Y$ $\quad f: X \to Y$
Ho-
Bijectn + BiCont

$f^{-1}: Y \to X$
Bijectn + Cont + f is Cont

$\Rightarrow X \approx Y \Leftrightarrow Y \approx X$

VU

3: $X \approx Y$, $Y \approx Z$

$f: X \to Y$ $\quad g: Y \to Z$

$g \circ f: X \to Z$

$\to g \circ f$ is continuous

$f^{-1} \circ g^{-1}$ is also continuous

$\Rightarrow X \approx Z$

Homeomorphism...

Define **Equivalence Relation**:
A binary relation $\sim$ on a set X is said to be an equivalence relation iff $\sim$ satisfies the following three properties for all $a, b, c \in X$

1. **Reflexive**: $a \sim a$
2. **Symmetric**: $a \sim b$ iff $b \sim a$
3. **Transitive**: If $a \sim b$ and $b \sim c$ then $a \sim c$

Topological Properties

Some Topological Properties:

- Cardinality of set X.
- Cardinality of $\mathcal{T}$.
- Connectedness of space X.
- Being Discrete.

Metric on a Set

Define **Metric:** on a set

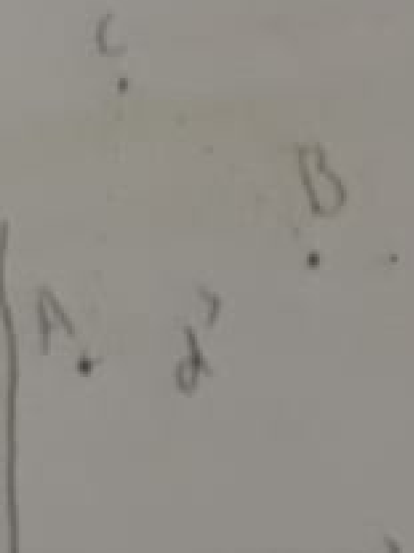
A metric on a non empty set X is a real valued function d defines on $X \times X$, i.e

$$d : X \times X \longrightarrow \mathbb{R}$$

such that for all $x, y, z \in X$, the following conditions are satisfied:

1. $d(x, y) \geq 0$, equality holds iff $x = y$
2. $d(x, y) = d(y, x)$ (Symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle Inequality)

Metric on a Set



A metric is a distance function i.e. MCQ

A function that defines a distance between each pair of elements of a set.

- Distance between A and B is always positive
- Dis bet A and A is "0"
- "0" is the dis only from A to A
- Dis bet A and B = Dis bet B and A
- dis bet A and B $\leq$ dis bet A and C + dis bet C and B

Topological Properties

$[0,2]$, $Y\ [0,4]$

, $L_Y = 4$

$\to Y$

$\to 2x$ Biject.

+ Cont

+ Inv. of f is also cont

Y

Some Non Topological Properties:

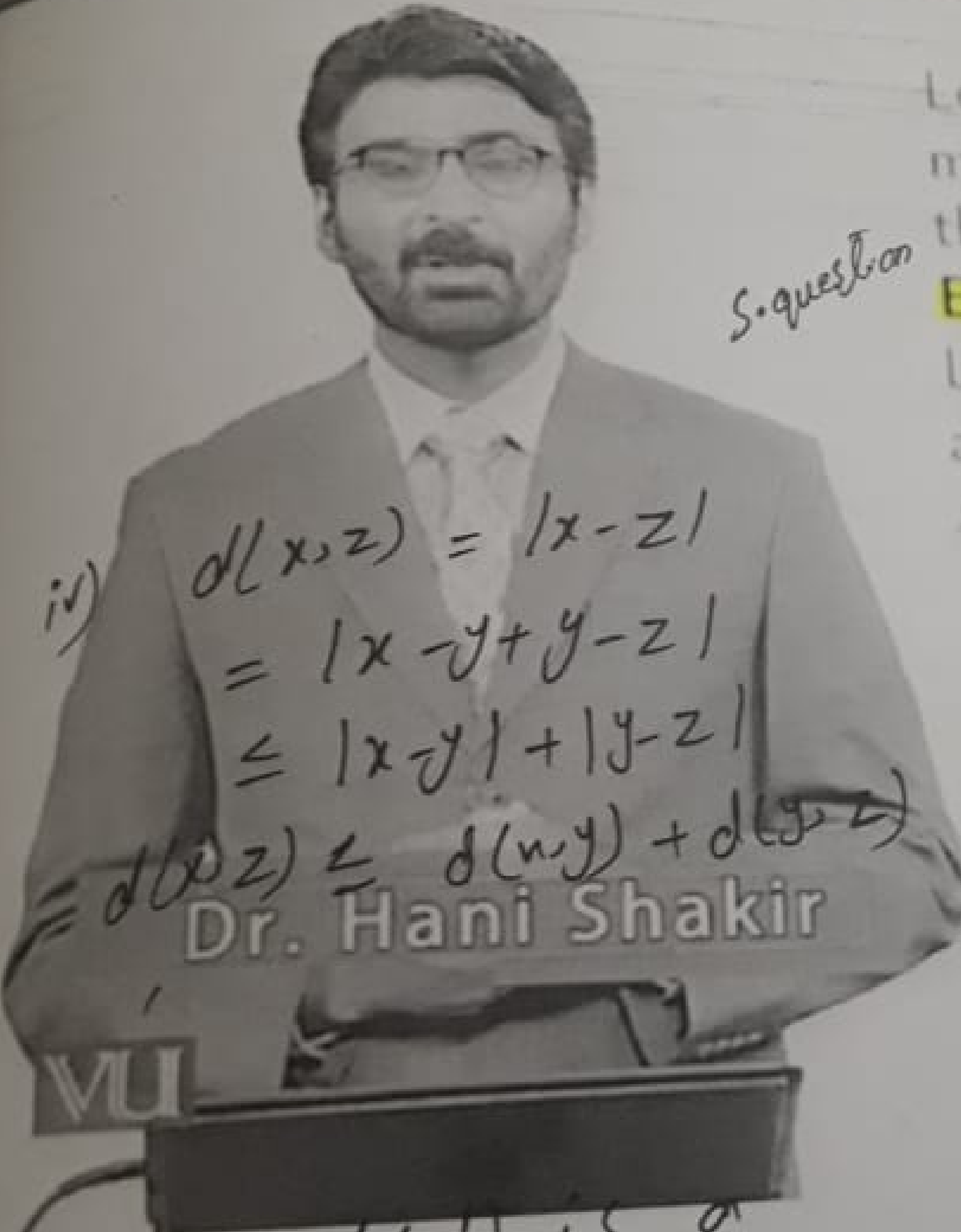
- Length
- Area
- Boundedness

Metric on a Set

Define **Metric Space**:

A set X with a metric d defined on it is called a metric space.

Examples of Metric



Let us see some examples of metrics and verify all the three axioms for each metric.

S.question **Example 1:**

Let $X = \mathbb{R}$ and d is defined as $d(x, y) = |x - y|$ where $x, y \in \mathbb{R}$.

Sol:

i) As $|x-y| \geq 0$ $\therefore d(x,y) \geq 0$

ii) $d(x,y) = |x-y|$ $\because |x| = |-x|$

$= |y-x|$

$= d(y,x)$

iii) $d(x,y) = 0 \iff |x-y| = 0$

$d(x,y) = 0 \iff x-y = 0$

$d(x,y) = 0 \iff x = y$

iv) $d(x,z) = |x-z|$

$= |x-y+y-z|$

$\leq |x-y| + |y-z|$

$\Rightarrow d(x,z) \leq d(x,y) + d(y,z)$

Then (X,d) is a metric space.

Usual Metric on $\mathbb{R}^n$

Usual Metric on $\mathbb{R}^n$

Consider $\mathbb{R}^n$ and d is defined as

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

where $x = (x_1, x_2, \cdots, x_n), y = (y_1, y_2, \cdots, y_n) \in \mathbb{R}^n$.

Usual Metric on $\mathbb{R}^n$

imp

Usual Metric on $\mathbb{R}^2$

Consider $\mathbb{R}^2$ and d is defined as

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

where $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.

Unusual Metrics on $\mathbb{R}^n$

Example 1:

Consider $\mathbb{R}^2$ and d_1 is defined as

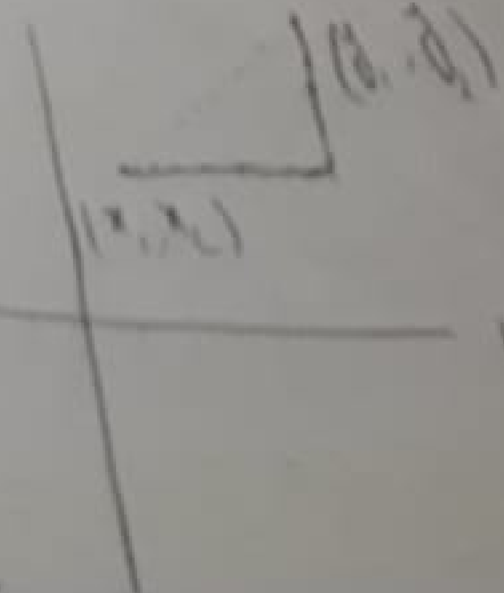
$$d_1(x, y) = \max(|x_1 - y_1|, |x_2 - y_2|)$$

where $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.

$x = (2, 3), y = (7, 15)$

$d_1(x, y) = \max(|2-7|, |3-15|)$

$= \max(5, 12) = 12$



Unusual Metrics on $\mathbb{R}^n$

Example 2:

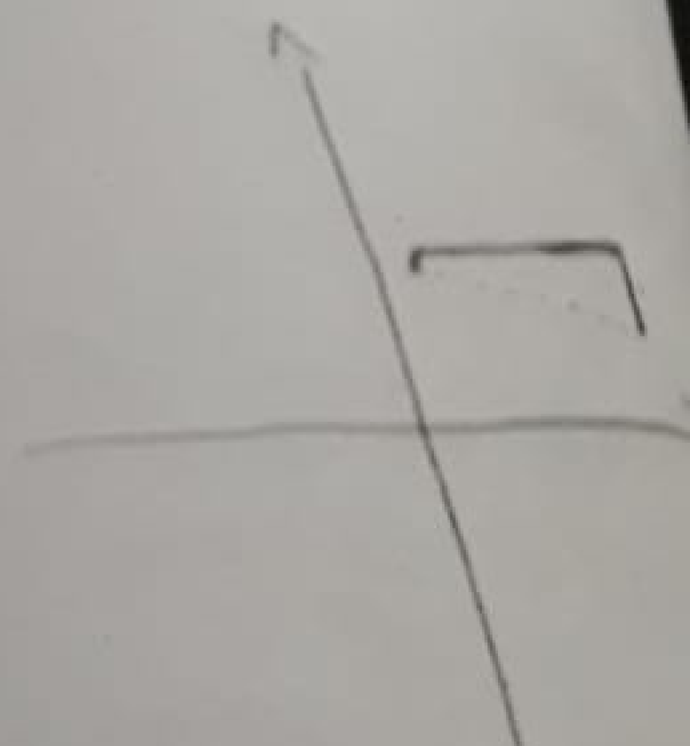
Consider $\mathbb{R}^2$ and d_2 is defined as

$$d_2(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

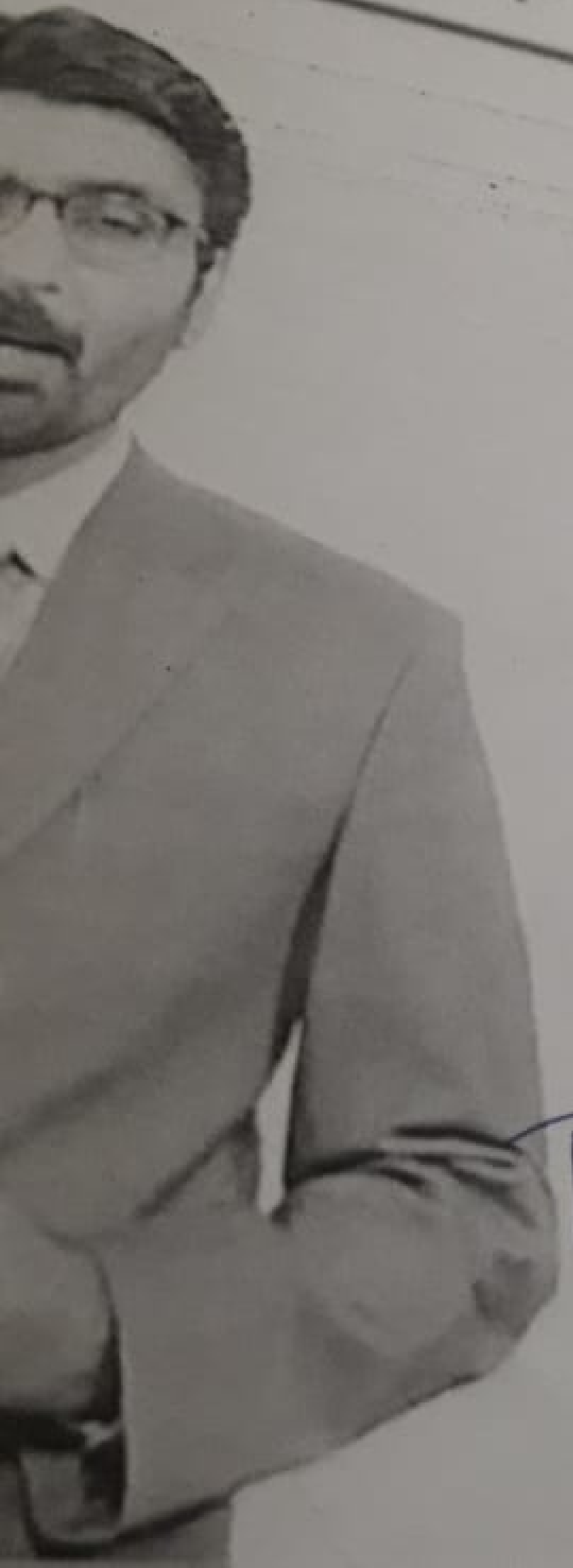
where $x = (x_1, x_2)$, $y = (y_1, y_2) \in \mathbb{R}^2$.

$x = (1, 3) \qquad y = (7, 2)$

$$d_2(x, y) = |1 - 7| + |3 - 2|$$
$$= 6 + 1 = 7$$



Open Ball



Define open ball

Consider X with a metric d. For any $x \in X$ and any real number $r > 0$, the set of all points $y \in X$ whose distance from x is less than r is called the **open ball** of radius r centered at x.

i.e.

imp

$$\boxed{B(x,r) = \{y \in X \mid d(x,y) < r\}}$$

Balls Centered at One Point

Fact:

Let $B_1(x, r_1)$ and $B_2(x, r_2)$ be two open balls centered at x with radius r_1 and r_2 respectively. Then either $B_1(x, r_1) \subset B_2(x, r_2)$ or $B_2(x, r_2) \subset B_1(x, r_1)$.

imp

Open Ball

Example:
Recall $\mathbb{R}$ with usual metric $d(x, y) = |x - y|$. Consider $x = 2$ and $r = 1$.
Let us see what is $B(2, 1)$?

$\boxed{B(2,1)} = \{y \in \mathbb{R} \mid d(2,y) = |2-y| < 1\} = (1, 3)$

$|2-y| < 1$

$-1 < 2-y < 1$

$\boxed{1 < y < 3}$

$|2-y| < 1$

$|y-2| < 1$

$-1 < y-2 < 1$

add 2:

$\Rightarrow -1+2 < y-2+2 < 1+2$

$\Rightarrow 1 < y < 3$

VU

$|x-y| < r$

$-r < x-y < r$

$-1 < 2-y < 1$

add 2:

$-1+2 < 2-y+2 <$ ...

$1 <$ ...

Examples: Metric Topology

Example 1:
Recall $\mathbb{R}$ with usual metric $d(x, y) = |x - y|$ where $x, y \in \mathbb{R}$.
Here $\mathcal{B} = \{B(x, r) | x \in X, r > 0\}$ is the set of all open intervals.

McQs

And we know that the topology generated by the set of all open intervals is usual topology.
So the metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is usual topology on $\mathbb{R}$.

VU

Metric Topology

Define **Metric Topology:**

Let X be a nonempty set with metric d. The topology $\mathcal{T}$ on X generated by the set of all open balls in X with respect to d is called the "Metric Topology."
We also say that a topology $\mathcal{T}$ on X induced by the metric d.

Define **Metric Space:**

Consider X with a metric d. this d induces a topology the "metric topology." (X, d) is called a metric space.

Examples: Metric Topology

Example 2:

Recall $\mathbb{R}^2$ with usual metric
$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$ where
$x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$
Here $\mathcal{B} = \{B(x, r) | x \in X, r > 0\}$ is the set of all open disks.

MCQs

And we know that the topology generated by the set of all open disks is usual topology on $\mathbb{R}^2$. So the metric topology induced by $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$ on $\mathbb{R}^2$ is the usual topology on $\mathbb{R}^2$.

Consider a non empty set X with a metric d defined on X. Now this metric induces a topology on X. Let's reverse this process, i.e.

We have a set X with a topology T, now the question is

Q: "Dose there exist a metric d such that d on X induces the topology T?"

Ans: The answer depends on the selection of T.

Example 3:
Recall trivial metric on a set X i.e

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

where $x, y \in X$. Here
$\mathcal{B} = \{B(x,r) | x \in X, r > 0\} = \{\{x\}, X | x \in X\}$

MCQs

And we know that the topology generated by the set of all single tons is discrete topology. So the metric topology induced by the above metric is the discrete topology on X

…rizable Spaces

Define **Metrizable Space**:

A topological space (X, T) is called metrizable space if there exists a metric d on X that induces the topology T.

Remark:

MCQs Not all spaces are metrizable.

Examples: Metrizable Spaces

Example 1:
Consider $\mathbb{R}$ with usual topology. Recall usual metric $d(x,y) = |x - y|$ on $\mathbb{R}$.
Now the topology induced by $d(x,y) = |x - y|$ on $\mathbb{R}$ is the usual topology on $\mathbb{R}$.
So $(\mathbb{R}, \mathcal{T}_u)$ is metrizable space.

Example 2:
Consider $\mathbb{R}^2$ with usual topology. Recall usual metric

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

on $\mathbb{R}^2$. Now the topology induced by d on $\mathbb{R}^2$ is the usual topology on $\mathbb{R}^2$. So $(\mathbb{R}^2, \mathcal{T}_u)$ is metrizable space.

VU

Non-Metrizable Spaces

Non-Metrizable space

Let us see some examples of non-metrizable spaces, i.e. the topological space $(X, \mathcal{T})$ such that there exists no metric which induces $\mathcal{T}$.

Examples: Non-Metrizable Spaces

Example 2:

Let X be an infinite set. Consider cofinite topology on X. Then it is not metrizable space.

Reason:

There is a topological property called "Hausdorffness" which should exist in a metric space. So if there exists some metric corresponds to $(X, \mathcal{T}_{cof})$ then $(X, \mathcal{T}_{cof})$ should be Hausdorff but it is not true.

So $(X, \mathcal{T}_{cof})$ is not metrizable space.

Caution:

It does not mean that if some topological space is Hausdorff then that space is metrizable.

VU

Examples: Non-Metrizable Spaces

Example 1:
Consider $X \neq \{x\}$ with indiscrete topology i.e $\mathcal{T}_{ind} = \{\emptyset, X\}$. Then it is not metrizable space.

Reason:
It is because of the fact that in a metric space all the finite set are closed. But here only closed sects are $\emptyset$ and X. So $\mathcal{T}_{ind}$ can not be induced by any metric.

Caution:
It does not mean that if in some topological space all finite sets are closed then that space is metrizable

VU

First Countable Spaces

Define 1^{st} countable space.

A topological space $(X, \mathcal{T})$ is called a "First Countable Space" if for each $p \in X$ there exists a countable local base at p.

⋮

First Countable Spaces

Remark:

Every subspace of a first countable space is first countable

$A \subset X$ (X is first countable)

$\Rightarrow$ A is also first countable.

ples: First Countable Spaces

Let us discuss some examples of first countable spaces

Example 1:

MCQs

Discrete space is first countable.

Remember a local base at a point p in discrete space, i.e.

$$\mathcal{B}_p = \{\{p\}\}$$

and it is finite.

Examples: First Countable

Example 4:

Every metric space is first countable

Sol: Let X be a metric spa

Now consider the following co

open balls for $p \in X$.

$$\mathcal{B}_p = \{ B(p, \tfrac{1}{n}) \mid n \in \mathbb{N}$$

$\mathcal{B}_p$ is a local base at p

first countab

Remark: Q. 2019 MTH634

Every subspace of a second countable space is second countable

Proof: Let $A \subset X$ (X is second countable space)

So $\exists$ $\mathcal{B} = \{B_n \mid n \in \mathbb{N}\}$ countable

Now $\mathcal{B}_A = \{A \cap B_n \mid n \in \mathbb{N}\}$ is a basis for $\mathcal{T}_A$ And $\mathcal{B}_A$ is countable

$\Rightarrow$ A is second countable $\square$

ples: Second Countable

Let us see discuss some examples of second countable spaces.

Example 1:

Let X be a countable set with discrete topology. Then (X, T_{Dis}) is second countable. Recall basis for discrete space, i.e.

$$\mathcal{B} = \{\{x\} | x \in X\}$$

Examples: Second Countable

Example 3: write one example of 2^{nd} countable?

MCQs + Short question

$\mathbb{R}$ with usual topology is second countable.

Consider the following basis for usual topology on $\mathbb{R}$.

$$B = \{(a, b) \mid a, b \in \mathbb{Q}\}$$

So $(\mathbb{R}, \mathfrak{J}_u)$ is second countable.

countable basis

Ex. 4 $(\mathbb{R}, \mathfrak{J}_{Dis})$ is not second countable.

$$B_{\mathfrak{J}_{Dis}} = \{\{x\} \mid x \in \mathbb{R}\}.$$

Examples: Second Countable

S. question

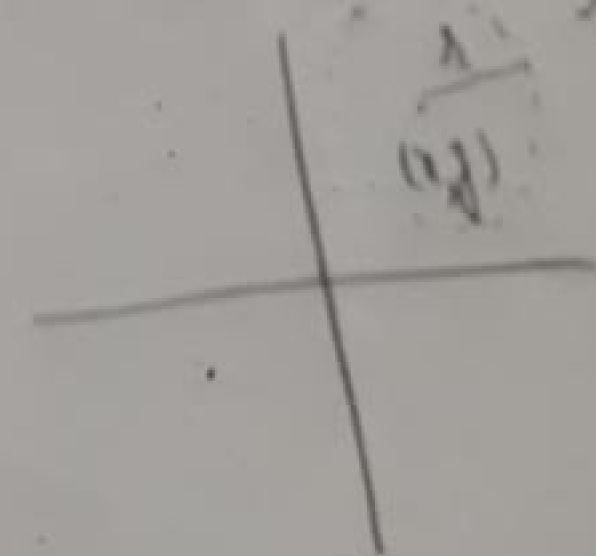
Example 5:

MCQ's

Show that $\mathbb{R}^2$ with usual topology is second countable

Consider a basis $\mathcal{B}$ for usual topology on $\mathbb{R}^2$ consists of collection of open disks with rational radii and centers with rational coordinates. Such collection is countable.

$r, x, y \in \mathbb{Q}$



- First Countability Axiom
- Second Countability Axiom

Theorem: MCQs

Let a topological space $(X, \mathcal{T})$ is second countable space. Then it also first countable space.

It has already been proved that every second countable space is first countable.

But what about its reveres implication, i.e.

For a first countable space what one can say about its second countability?

Before answering this question, let us first discuss

First Countability $\not\Rightarrow \ldots$

MCQs + short questions

Example 2:

Consider $\mathbb{R}$ with usual topology. It is both first countable and second countable.

Example 3:

Consider $\mathbb{R}$ with lower limit topology. It is first countable but not second countable.

Remark:

A first countable space may or may not be second countable.

Open Cover

Define Cover:(In Topology)?

Let $(X, \mathcal{T})$ be topological space. A cover of a X is a collection of subsets of X, i.e.

$$\mathcal{U} = \{U_i | U_i \subset X, i \in I\}$$

where I is index set, such that

$$X = \bigcup_{i \in I} U_i$$

Open Cover

$X = [0,1]$

Define Cover: ?

A cover of a set X is a collection of sets

$$\mathcal{U} = \{U_i \mid i \in I\}$$

where I is index set, such that

$$X \subset \bigcup_{i \in I} U_i$$

2] }

2] $\mathcal{U}$ is cov

$\not$ [0,1]

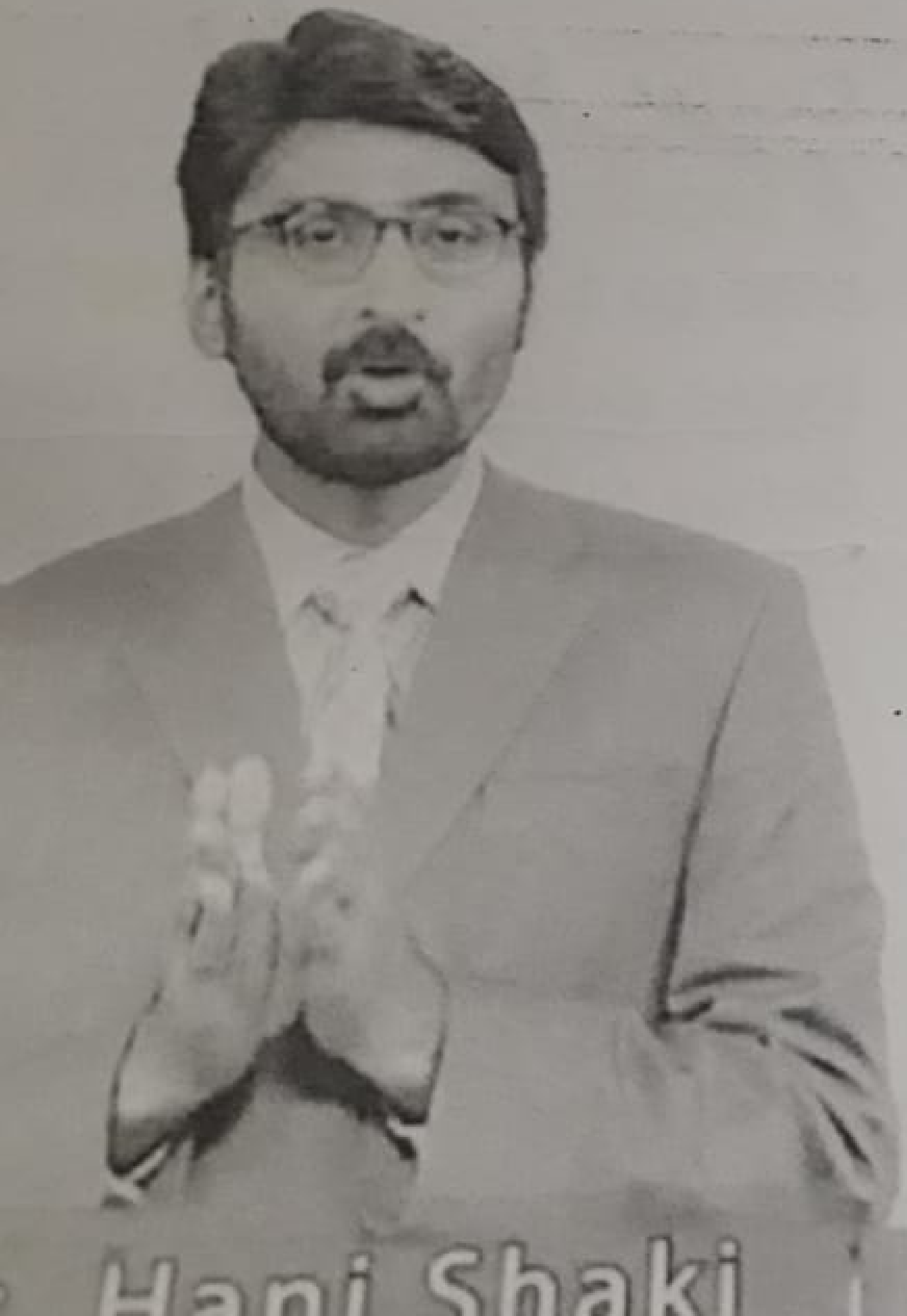
1.9], (1.8, 3)}

Lindelöf Space

Define Lindelöf Space:?

A topological space (X, T) is called a Lindelöf space if and only if every open cover of X has a countable subcover.

Second Countable...



Theorem: MCQs

Every second countable space is Lindelöf space.

Lindelöf Space

Remark: MCQs

A subspace of a Lindelöf space needs not be Lindelöf

Separable Spaces

Define separable?

A topological space $(X, \mathcal{T})$ is said to be "Separable" if there exists a countable dense subset A of X.

i.e.

$\exists \; A \subset X$ such that

1. A is countable.
2. $\overline{A} = X$.

Separable Spaces

Remark: MCQs

Every second countable space is separable. The converse of this is not true in general.

Counter Example: MQs

$\mathbb{R}$ with lower limit topology is separable but not second countable.

Examples: Separable Spaces

Let us discuss some examples of separable spaces.

Example 1:
Let X be a finite set. Then X with any topology is separable.

Example 2:
Consider a set X with indiscrete topology. Then X separable.

Some More Examples

Proposition 1: long

Show that A discrete space is separable if and only if it is countable

Proof. X, $\mathfrak{I}_{dis} = P(X)$

We know that the only dense subset in X is X itself.

So the only choice for A is X s.t $\overline{A} = X$

$\Rightarrow$ X is separable iff X is countable.

VU

and separability of metric spaces. In fact a metric space is not second countable and not separable in general.

Example:

An infinite set X with trivial metric is not second countable and not separable.

Metric Spaces and Separability

Theorem:
A separable metric space is second countable

Proof:

Separation Axioms

1. T_0 Spaces
2. T_1 Spaces
3. T_2 (Hausdorff) Spaces
4. Regular Spaces
5. T_3 Spaces
6. Normal Spaces
7. T_4 Spaces

T_0 Space

short question

Define To space

A topological space (X, T) is said to be "T_0 Space" iff for each $x, y \in X$ such that $x \neq y$ there exists an open set $U_x \subset X$ containing x such that $y \notin U_x$

or

there exists an open set $V_y \subset X$ containing y such that $x \notin V_y$.

T_1 Space

MTH-634

Def:

A topological space (X, T) is said to be "T_1 Space" iff for each $x, y \in X$ such that $x \neq y$ there exist open subsets U_x, U_y of X containing x, y respectively such that $y \notin U_x$ and $x \notin U_y$

Q. 2019

Examples: T_1 Space

show that long

Example 3: Every cofinite space is T_1

Sol: Let $(X, \mathfrak{J}_{cof})$

Let X is finite

$\mathfrak{J}_{cof} = P(X)$ discrete

If X is not finite Let $U \in \mathfrak{J}_{cof}$

iff $U = \phi$ or U^c = finite

$x, y \in X$ $x \in \underset{\not\ni y}{U_x} = X \setminus \{y\}$; $y \in \underset{\not\ni x}{U_y} = X \setminus \{x\}$

VU

Example: A T_0 space that is not T_1.
Consider $\mathbb{R}$ with $\mathcal{T}$ generated by

$$\{(a, \infty) \mid a \in \mathbb{R}\}$$

Prove That every T_1-space is T_0-space. long

Proof:
Let X be T_1-space. To prove X is T_0-space.
Let $x, y \in X$. such That $x \neq y$. Since X is T_1 so There exists two open sets U and V in X such That $x \in U$, $y \notin U$. and $y \in V$, $x \notin V$.
Since here we have an open set U in X. with $x \in U$, $y \notin U \Rightarrow X$ is also T_0-space.

lecture:
open cover
sub-cover

Let us discuss some examples of T_1 spaces.

Example 1:

Every discrete space is T_1.

Example 2:

Every metric space is T_1.

Some More Exam

Proposition 2:
$(X, \mathcal{T}_{cof})$ is separable

Examples: Separable Spaces

Example 3:

$\mathbb{R}$ with usual topology is separable.

Reason:

Since there exists $\mathbb{Q}$ in $\mathbb{R}$ that is countable and

$$\overline{\mathbb{Q}} = \mathbb{R}.$$

Example 4:

$\mathbb{R}$ with discrete topology is not separable.

Reason:

Since there exists no countable subset in $\mathbb{R}$ that is den

$\mathbb{R}$.

Lindelöf Space

MCQs + Short questions

Example 1:

Let X be a finite set with any topology then it is Lindelöf

Example 2:

A set X with indiscrete topology then it is Lindelöf

Example 3:

$X = \mathbb{R}$ with usual topology is Lindelöf

Examples: Non-Metrizable Spaces

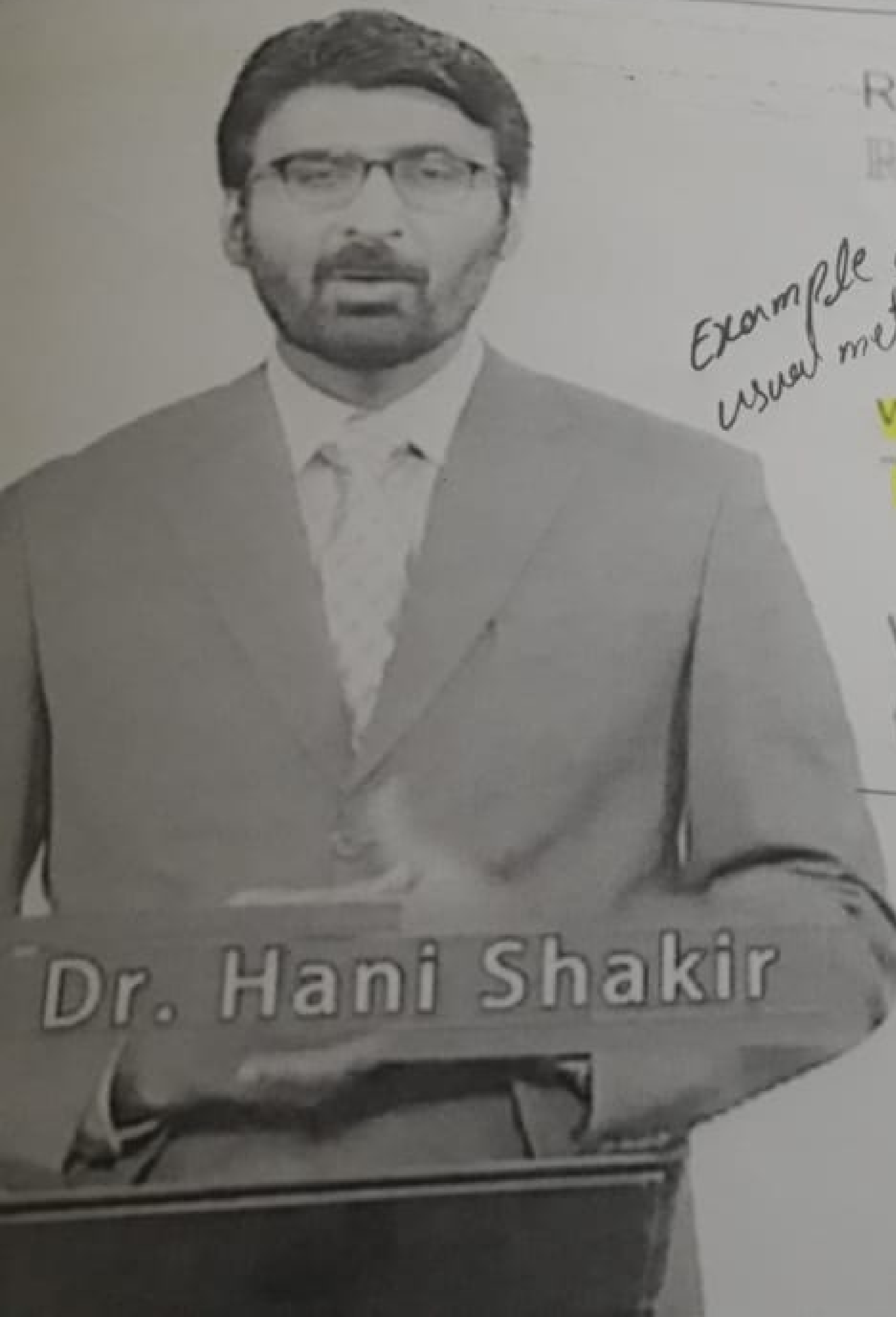
Example 3:

$\mathbb{R}$ with lower limit topology is not metrizable space.

Reason:

There are topological spaces called "Second Countable Spaces" and separable metric spaces are second countable. So if there exists some metric corresponds to $(\mathbb{R}, \mathcal{T}_l)$ then $(\mathbb{R}, \mathcal{T}_l)$ should be second countable but it is not true. So $(\mathbb{R}, \mathcal{T}_l)$ is not metrizable space.

Usual Metric on $\mathbb{R}^n$



Recall the metric defined on $\mathbb{R}$ in "Example 1"

Example of usual metric on $\mathbb{R}$

$$d(x, y) = |x - y|$$

where $x, y \in \mathbb{R}$.

This is the usual metric in $\mathbb{R}$.

We can also write it in the following way

$$d(x, y) = \sqrt{(x - y)^2}$$

uity and Open Mapping

A map $f : X \to Y$ is a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. f is bijective.
2. f is continuous.
3. f is open map.

Projection Mapping

Projection mapping is an example of mapping that is not closed in general

But let us see first
"What is projection map?"

MTH 634

Module 86 To 132
for Final Term
Created by Badar Toor
MSc Math Bahawalnagar
fb.com/badartoor

128 → 140

172 → 183